

The general- λ Gamma/log asymptotic

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Abstract

Step 3 of the general- d equal-ratio monomial milestone (Astra #14). The one-dimensional integral produced by the product density of unit 172 satisfies

$$\int_0^1 z^{\lambda-1} (-\log z)^m e^{-\beta N z} dz \sim \Gamma(\lambda) \beta^{-\lambda} N^{-\lambda} (\log N)^m \quad (N \rightarrow \infty),$$

for every $\lambda > 0$, $\beta > 0$, $m \in \mathbb{N}$ (tendsto_gammaLogIntegral_div, gammaLogIntegral_isEquivalent). A Lebesgue-integral bridge (lintegral_logDensity_mul_exp) connects the real-valued statement to the $[0, \infty]$ -valued product-density integrand. Zero sorry/axiom.

1 The things formalised

```
def gammaLogIntegral (l : ℝ) (m : ℕ) (N : ℝ) : ℝ :=
  z in Ioc 0 1, logDensity l m z * Real.exp (-( * N * z))
def logTailIntegral (l : ℝ) (i : ℕ) (N : ℝ) : ℝ :=
  t in Ioc 0 N, t ^ (l - 1) * (-Real.log t) ^ i * Real.exp (-( * t))
theorem integral_rpow_mul_exp_neg_mul (hl : 0 < l) (h : 0 < ) :
  t in Ioi 0, t ^ (l - 1) * Real.exp (-( * t)) = Real.Gamma l * ^ (-1)
theorem abs_logTailIntegral_le ... : |logTailIntegral l i N|
  t in Ioi 0, |Real.log t| ^ i * t ^ (l - 1) * Real.exp (-( * t))
theorem gammaLogIntegral_eq_scaled (hN : 0 < N) : gammaLogIntegral l m N =
  N ^ (-1) * t in Ioc 0 N, t ^ (l-1) * (log N - log t) ^ m * exp (-( * t))
theorem scaled_integral_eq_sum ... = ∑ j range (m+1),
  log N ^ j * (m.choose j) * logTailIntegral l (m - j) N
theorem tendsto_gammaLogIntegral_div (hl : 0 < l) (h : 0 < ) :
  Tendsto (fun N => gammaLogIntegral l m N / (N ^ (-1) * Real.log N ^ m))
  atTop ( (Real.Gamma l * ^ (-1)))
theorem gammaLogIntegral_isEquivalent (hl : 0 < l) (h : 0 < ) :
  (fun N => gammaLogIntegral l m N) ~[atTop] powLog (Real.Gamma l * ^ (-1)) l m
theorem lintegral_logDensity_mul_exp (hl : 0 < l) (hN : 0 < * N) :
  z in Ioc 0 1, ofReal (logDensity l m z) * ofReal (exp (-( * N * z)))
  = ofReal (gammaLogIntegral l m N)
```

2 Mathematics

Substituting $t = Nz$ gives $N^{-\lambda} \int_0^N t^{\lambda-1} (\log N - \log t)^m e^{-\beta t} dt$. Expanding $(\log N - \log t)^m = \sum_j \binom{m}{j} (\log N)^j (-\log t)^{m-j}$ and dividing by $N^{-\lambda} (\log N)^m$ leaves $\sum_j \binom{m}{j} (\log N)^{j-m} J_{m-j}(N)$ with $J_i(N) = \int_0^N t^{\lambda-1} (-\log t)^i e^{-\beta t} dt$. Each J_i is bounded uniformly in N by $\int_0^\infty |\log t|^i t^{\lambda-1} e^{-\beta t} dt <$

∞ , so every term with $j < m$ tends to zero, while $J_0(N) \rightarrow \int_0^\infty t^{\lambda-1} e^{-\beta t} dt = \Gamma(\lambda)\beta^{-\lambda}$. The plan recorded after unit 172 was dominated convergence; the binomial split is shorter because the envelope integrability was already in the seabed (`log_pow_fluctuation_integrableOn` with $a = 0$), and it produces the bounded-tail lemma as a reusable by-product.

3 Paper-facing meaning

Combined with unit 172 this gives, for the equal-ratio monomial $\prod_i x_i^{2k_i}$ with $(h_i + 1)/(2k_i) = \lambda$ in d variables, the leading behaviour $N^{-\lambda}(\log N)^{d-1}$ with explicit constant $\Gamma(\lambda)\beta^{-\lambda}/(d-1)!$ times the Jacobian factors $\prod_i 1/(2k_i)$ from unit 170. Remaining: the Fin d box bridge (unit 174) and the assembly (unit 175).

4 Lean notes

- `rw [sub_eq_add_neg]` rewrites the *first* subtraction, which here was the exponent $\lambda - 1$, not $\log N - \log t$; use an explicit `show ... by ring` rewrite.
- `pow_sub` produces $a^m \cdot (a^n)^{-1}$, not a division; close with `mul_inv, inv_inv, div_eq_mul_inv, mul_comm`.
- `Asymptotics.isEquivalent_of_tendsto_one` on this pin takes only the ratio limit (no eventual-vanishing side condition); the `~[1]` notation needs `open Asymptotics`.
- `integral_finset_sum/tendsto_finset_sum` are deprecated for `integral_finsetSum/tendsto_finsetSum`.
- `Tendsto.zero_mul_isBoundedUnder_le` with `isBoundedUnder_of` accepts an `||·||` `M` bound directly (`Real.norm_eq_abs` is definitional).